PROMPT FOR RESOLVING THE DEGREE-57 MOORE GRAPH

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Abstract. This document contains a full frontier-model research prompt for deciding whether the last possible unknown diameter-2 Moore graph exists. The target is a strongly regular graph with parameters $(3250, 57, 0, 1)$, or a complete proof of nonexistence. Status snapshot: 20 July 2026.

1. Prompt

Current task statement

A Moore graph of degree k and diameter 2 is a k-regular simple graph of diameter 2 and girth 5 attaining the Moore bound $1 + k + k(k-1) = k^2 + 1$ vertices.

Resolve the degree-57 case completely. Such a graph would have 3250 vertices and is equivalently a strongly regular graph with parameters

$$(v, k, \lambda, \mu) = (3250, 57, 0, 1).$$

For its adjacency matrix A, the defining equations are

$$A\mathbf{1} = 57\mathbf{1}, \qquad A^2 = 56I - A + J.$$

Assume for purposes of this task that a complete resolution is reachable. Do not assume existence or nonexistence.

Complete-resolution standard

A complete resolution must prove one alternative:

- Positive alternative: give a simple graph on 3250 vertices satisfying the matrix equations exactly;
- Negative alternative: prove that no such graph exists, with a complete conceptual argument or a certified exhaustive computation;
- in either direction, justify every spectral, group-theoretic, local-tree, and automorphism reduction without assuming symmetry that an arbitrary graph need not possess.

What does not count

Partial progress is insufficient unless it logically implies the exact resolution above. In particular, the following do not count:

- a graph with the correct degree and diameter but a triangle or 4-cycle;
- a feasible spectrum, intersection array, or partial association scheme without an actual graph;
- excluding only vertex-transitive, Cayley, cyclic, or nontrivially symmetric candidates;
- a local completion around one vertex that has not been globally closed consistently;
- a numerical or heuristic failure to construct the graph;
- a SAT or algebraic infeasibility result for a restricted ansatz without proving the ansatz is universal.

Use multiagent v2 aggressively and dynamically. You have up to 64 concurrent agents available. Do not use a fixed assignment such as "N agents for strategy X." Instead, manage the search using the following heuristics:

- Begin with a genuinely diverse portfolio of approaches. Agents should explore substantially different formulations, invariants, reductions, algebraic viewpoints, structural inductions, decompositions, optimization methods, exact computational searches, and formal-proof routes.
- Do not tell most agents the currently favored approach. Preserve independence during early rounds so that agents do not all converge to the same attractive but incomplete reduction.
- Maintain an explicit registry of approach families. Group agents by the mathematical mechanism they are using, not by superficial wording. If too many agents converge to one family, redirect some toward underexplored formulations.
- Do not allow one route to dominate merely because it gives elegant reductions. A route that ends at a missing lemma equivalent in strength to the original problem is not close to completion unless it supplies a genuinely new proof of that lemma.
- When an approach stalls at a theorem-strength missing lemma, mark that route as blocked. Reopen it only when an agent proposes a materially new mechanism, invariant, construction, or exact experiment.
- Keep several incompatible proof or construction routes alive through multiple rounds. Cross-pollinate ideas only after independent agents have developed them far enough to expose their real strengths and gaps.
- Use adversarial agents throughout. Every candidate theorem, construction, certificate, reduction, and program must be attacked independently. Assign separate agents to mathematical correctness, completeness of case analysis, numerical exactness, implementation soundness, and novelty checking.
- Require agents to return concrete lemmas, constructions, equations, counterexamples to proposed sublemmas, executable encodings, or formally stated proof obligations. Reject status reports, vague optimism, and claims that a global compatibility step is "routine."
- The root agent should repeatedly synthesize, challenge, redirect, and launch new rounds. Do not stop after the first wave fails. Convert promising patterns into exact conjectures and send those conjectures to independent proof and falsification teams.

Problem-specific search portfolio

In the initial independent rounds, ensure that the portfolio includes, but is not limited to, the following genuinely different families:

- fix a root and use the forced depth-two breadth-first tree: 57 neighbors and $57 \cdot 56 = 3192$ distance-two vertices, then encode all branch-to-branch matchings;
- permutation and transition-system formulations for edges among the 56-sets in distinct branches, with exact compatibility equations;
- the strongly regular graph algebra, spectrum $57, 7, -8$, star complements, integral representations, modular eigenspaces, and p-adic constraints;
- triple-intersection numbers, Krein parameters, subconstituents, coherent configurations, and association-scheme refinements;
- automorphism-group restrictions, including the absence of involutions, used only as case reductions rather than universal symmetry assumptions;
- SAT, SMT, CP-SAT, exact cover, and algebraic constraint solving on canonical partial completions, with proof-producing unsatisfiability;
- coding-theoretic, lattice, spherical-embedding, design-theoretic, and finite-geometric constructions or obstructions;
- new global counting identities that couple multiple rooted neighborhoods and force an impossible parity, rank, or intersection pattern.

Mandatory adversarial audit

Any surviving candidate must be checked against every item below by agents that did not originate the candidate:

- for a candidate graph, verify symmetry, zero diagonal, binary entries, all row sums 57, and $A^2 = 56I - A + J$ exactly;
- independently verify diameter 2, absence of triangles and 4-cycles, and uniqueness of the common neighbor of every nonadjacent pair;
- check the forced spectrum and multiplicities as redundant invariants;
- audit all branch labels, permutation encodings, and local-to-global consistency conditions for

omitted edge types;

- do not infer universal automorphisms from a symmetric search construction; retain and certify the automorphism-free cases;
- if nonexistence is computational, prove the canonical generation is exhaustive and validate every proof certificate independently;
- test all code on the known Moore graphs of degrees 2, 3, and 7 before trusting the degree-57 search.

Required final artifact

The final response must provide the complete mathematical resolution and its finite certificate:

- for existence, a compact adjacency representation, hashes, and two independent exact verifiers;
- for nonexistence, the complete structural reduction, case database, proof certificates, checkers, and reproducibility manifest;
- a human-readable derivation of the matrix equation and the local branch constraints;
- all source code, commands, software versions, expected counts, and intermediate hashes needed to rerun the audit.

Public search may be used for standard Moore-graph theory, known spectral and automorphism restrictions, and previous construction attempts. Do not search for a claimed secret graph or merely answer that the degree-57 case is open.

Do not return merely because current approaches fail or agents report theorem-strength gaps. Continue launching new rounds, reopening blocked approaches only when there is a genuinely new mechanism, and searching for fresh formulations.

Return only when a complete resolution has been found and survives adversarial audit. Do not return a reduction, a special case, an isolated missing lemma, an improved bound that leaves the target unresolved, a "best effort" literature review, or an explanation of why the problem is difficult.

Spend at least 8 hours of sustained search before even considering return or termination. If the execution environment imposes a hard stop before a complete resolution, do not fabricate success or elapsed work: preserve all machine-checkable artifacts and report only the strongest rigorously verified derivation together with the exact remaining gap.